

EVGrid AI

Smart EV Charging Network Planning System for Kopargaon Taluka

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1. PROJECT OVERVIEW

Project Name

EVGrid AI is an AI- and GIS-powered decision support system for scientific planning of EV charging infrastructure in Kopargaon Taluka. The source report identifies the region as a gateway to Shirdi with tourist, commuter and highway movement.

Problem Statement

The rapid adoption of electric vehicles requires strategically planned charging infrastructure. However, identifying suitable locations based on demand, accessibility, and grid availability remains challenging. Develop a planning system that assists authorities in designing an efficient EV charging network.

Proposed Solution

EVGrid AI combines GIS mapping, weighted site-suitability scoring, demand forecasting, grid-feasibility analysis and budget planning. The system ranks candidate locations and provides government authorities with a consolidated planning view and decision report.

Brief Technical Overview

- **Spatial layer:** GIS-based visualization of roads, villages, facilities, power infrastructure and existing charging stations.
- **AI layer:** Multi-parameter suitability scoring and future EV-demand forecasting.
- **Planning layer:** Location ranking, charger selection/count planning and budget prioritization.
- **Decision layer:** Dashboard, analytics and report generation for government planning.

2. SYSTEM ARCHITECTURE & TECHNOLOGY STACK

Architecture / Block Diagram

Government Officer / Planner → Planning Dashboard → GIS Engine + AI Engine + Analytics → Backend Services → PostgreSQL + PostGIS spatial database.

Key Modules

Module	Technical Function
GIS Engine	Spatial data processing, layer management, route/proximity analysis and location visualization.
AI Engine	Site suitability analysis, demand forecasting, scoring/ranking and recommendation generation.
Analytics	Performance matrices, trends, predictive insights and planning KPIs.
Backend Services	Application logic, APIs and data processing between dashboard and database.
Planning Dashboard	Government-facing monitoring, analysis, recommendations and report generation.

Technology Stack

Frontend: React.js | Backend: Node.js + Express.js | AI: Python/ML | GIS: Leaflet/QGIS | Database: PostgreSQL + PostGIS | External: Google Maps API | Auth: Firebase/RBAS | Version control: GitHub
The design targets a web-based government dashboard, secured APIs and spatial database services.

3. SYSTEM DESIGN

Database / ER-Oriented Design

Entity / Table	Key Fields / Purpose
Locations	Location ID, village name, latitude, longitude, population, traffic density, parking capacity, tourist score, commercial score.
Charging Stations	Station ID, name, latitude, longitude, charger type, capacity and status.
Electricity Grid	Substation name, transformer capacity, available power and grid load.
Demand Forecast	Year, predicted EVs, charging demand, peak hour and suggested chargers.

Data Flow

Input data	Processing	Output
Traffic, population, tourism, roads, grid	Car parking, EV + spatial mapping	GIS layers / candidate locations
Candidate locations + weighted factors	Suitability scoring + ranking	Excellent / Good / Moderate / Poor
Historical / future EV indicators	Demand Forecasting	Projected demand and suggested chargers
Scores + demand + budget	Planning optimization	Priority hubs + decision report

Logical Relationships

- A Location is evaluated against surrounding traffic, population, parking, tourism, commercial activity and grid context.
- A recommended location can be associated with a proposed Charging Station configuration.
- Electricity Grid records provide capacity/proximity inputs for feasibility scoring.
- Demand Forecast outputs support charger-count & future expansion decisions.

User Roles & Access Control

Role	Access
Government Officer / Planner	Dashboard, GIS analysis, recommendations, analytics and report generation.
System Administrator	User/security administration and platform maintenance (role implied by RBAC requirement).

4. TECHNICAL WORKFLOW & METHODOLOGY

Core Logic – Suitability Scoring

The project uses a weighted multi-parameter scoring model. Each candidate location receives normalized factor scores, combined using the documented weights.

Parameter	Weight
Traffic Density	25%
Population Density	20%
Grid Density & Proximity	20%
Parking Availability	10%
Commercial Area Activity	10%
Tourist / Pilgrim Flow	10%
Future EV Growth Potential	5%

Score Categories

Score	Decision
80–100	Excellent – Immediate Priority
60–79	Good – Phase 1/2 Deployment
40–59	Moderate – Secondary Phase
Below 40	Poor – Not Recommended

Planning Output Logic

Data → GIS layers → candidate locations → weighted scoring → ranking → demand/grid feasibility → budget → prioritization → final decision report

End-to-End Workflow

Data Collection → GIS Mapping → AI Analysis → Location Recommendation → Budget Planning → Final Decision Report

5. IMPLEMENTATION

Module-wise Implementation

Module	Implementation Focus
Data ingestion	Collect traffic, population, EV usage, POIs, tourism and infrastructure inputs; maintain spatially referenced records.
GIS mapping	Map records, villages, facilities, substations and existing charging stations; calculate proximate and spatial relationships.
AI recommendation	Normalize factor values, apply documented weighted factors, calculate suitability score and rank candidates.
Demand forecasting	Generate future demand estimates for 2027, 2030 and 2035 and translate demand into charger requirements.
Grid feasibility	Use substations/transform capacity, available power and grid load as planning constraints.
Budget optimizer	Prioritize locations and charger configurations within specified budget ranges.
Dashboard & Reporting	Present maps, scores, analytics, recommendations and downloadable decision reports.

Key Technical Components

Component	Documented Technology / Approach
Frontend	React.js-based responsive dashboard
Backend	Node.js + Express.js application services; Python components for AI/analytics
Database	PostgreSQL + PostGIS spatial storage
GIS	Leaflet / QGIS spatial visualization and analysis
External services	Google Maps API for location, route, distance and navigation functions
Authentication	Firebase Authentication / secure role-based access approach
Version control	GitHub repository and project management

Important Implementation Details

- The system is designed as a modular architecture so GIS, AI, analytics and reporting components can evolve independently.
- Spatial records use latitude/longitude and PostGIS-oriented storage for geographic queries.
- Recommendation results are explainable through the weighted factors and category thresholds rather than being a black-box single prediction.
- The dashboard is intended for government decision-makers, not as a consumer-facing charging-finder application.

6. TESTING & VALIDATION

Testing Approach

The technical validation approach is organized around data correctness, scoring correctness, GIS behavior, forecast output structure, budget constraints, access control and dashboard usability. The supplied project documents the architecture and intended behavior; it does not provide measured benchmark values or a completed test-case log. Therefore, the section records validation criteria without inventing numerical results.

Test Cases

ID	Test Case	Expected Validation
T01	Verify required location fields	Latitude/longitude and planning factors are available before scoring.
T02	Validate score weights	All documents weighted sum of 100% & contribute according to their defined percentages.
T03	Boundary score classification	Scores 80, 60, 40, and below 40 map to the documented categories.
T04	GIS candidate visualization	Candidate locations & relevant infrastructure layers appear in correct spatial context.
T05	Grid feasibility input	Substations capacity, available power & grid load are available for feasibility assessment.
T06	Forecast output schema	Forecast produces year, predicted EVs, charging demand, peak hour and suggested chargers.
T07	Budget prioritization	Recommended deployment respects the selected budget range and prioritizes higher-suitability sites.
T08	Role access	Restricted dashboard operations are protected through RBAC/security controls.

Validation Method

- **Functional validation:** Execute each workflow stage with representative candidate locations and inspect outputs.
- **Spatial validation:** Compare map layers, route/proximity relationships and selected hubs against the input geography.
- **Algorithm validation:** Use hand-calculated sample scores to verify implementation against the weighted formula.
- **Boundary validation:** Test scores around 40, 60 and 80 to confirm category transitions.
- **Security validation:** Verify role restrictions, password protection and secure API access.
- **Performance validation:** Measure dashboard load and map rendering against the documented targets.

Performance / Evaluation Metrics

Metric	Target from Project Design
Dashboard loading time	< 3 seconds
Map rendering	< 5 seconds
Operational uptime	99%
Scalability	Taluka → District → State

7. RESULTS, LIMITATIONS & FUTURE SCOPE

Results / Outputs

Output	Documented Result
Priority locations	Five proposed hubs: Kopargaon Railway Station, MSRTC Bus Stand, Main Commercial Market, Shirdi Highway Junction and Hospitals
Charger planning	Fast DC, fast + slow, slow AC and high-speed fast-charging configurations are proposed according to location use.
Budget ranges	Prototype/modeling: ₹3,000–₹10,000; pilot deployment: ₹25–40 lakh; Taluka-wide platform: ₹90 lakh–₹1.2 crore; full infrastructure: ₹5 crore–₹6 crore
Decision outputs	Suitability ranking, demand forecast, grid feasibility, budget prioritization, maps and final decision report.

Proposed Priority Hubs

Location	Purpose / Charger Configuration
Kopargaon Railway Station	Fast DC; tourist vehicles, taxis and available parking.
MSRTC Bus Stand	Fast + slow; buses, commuters and commercial vehicles.
Main Commercial Market	Slow AC; shoppers, commercial vehicles and long-stay parking.
Shirdi Highway Junction	High-speed fast charging; highway traffic and pilgrims.
Hospital Area	Emergency charging; EV ambulances and emergency public access.

Technical Limitations

- Data quality and freshness can affect recommendations; the design assumes periodic surveys and continuous data/API synchronization.
- Grid capacity may vary with peak demand and requires operational data for reliable feasibility decisions.
- Forecast quality depends on the availability and quality of historical EV/adoption data.
- The source report does not provide quantitative model-accuracy results, field-test statistics or production-scale performance measurements.

Future Technical Improvements

- Real-time IoT monitoring of charging-station status.
- Dynamic MSEDCL grid balancing and improved load management.
- Dynamic charging-slot reservation.
- Vehicle-to-grid (V2G) power sharing.
- Predictive maintenance for charging hardware.
- Expanded deployment from Kopargaon Taluka to district/state-scale planning.

TECHNICAL ARCHITECTURE & SYSTEM COMPONENTS

System Architecture

The architecture connects government officers to an interactive web dashboard powered by GIS, AI, and analytics engines built over a PostgreSQL/PostGIS spatial database.

Government Officer / Planner → Web Dashboard → GIS Engine + AI Engine + Analytics → Backend Server → PostgreSQL + PostGIS Database (roads, population, grid, traffic, EV and spatial data).

Technology Stack Architecture

Technology	Use in EVGrid AI
React.js	Frontend dashboard and responsive user interface.
Node.js + Express.js	Backend application services and APIs.
Python / ML	AI recommendation and forecasting components.
PostgreSQL + PostGIS	Relational and spatial data management.
Leaflet / QGIS	GIS visualization and spatial analysis.
Google Maps API	Location, route, distance and navigation services.
Firebase Authentication	Authentication/security layer identified in the original stack.
GitHub	Version control and repository management.

Functional Technical Flow

Input data → Spatial mapping → Candidate generation → Suitability scoring → Ranking → Demand forecasting → Grid feasibility → Budget prioritization → Decision report

Government Decision Outputs

- GIS maps and candidate locations
- Suitability ranking and score category
- Charging demand forecast
- Grid feasibility assessment
- Budget-prioritized deployment
- Final decision report

Technical Scope

The initial deployment target is Kopergaon Taluka. The architecture is modular and intended to scale to district, state and smart-city deployments without changing the core planning workflow.

8. REFERENCES

Technical References & Data Sources

1. Government of India — National Electric Mobility Mission Plan (NEMMP).
2. Government of Maharashtra — Maharashtra Electric Vehicle Policy 2021.
3. Ministry of Heavy Industries — EV initiatives and FAME India Scheme.
4. MSEDCL — Grid and power data.
5. NITI Aayog — Sustainable transportation and mobility reports.
6. OpenStreetMap (OSM) — Geospatial mapping data.
7. Ministry of Road Transport and Highways (MoRTH) — Road and traffic statistics.
8. Government of Maharashtra — Smart City initiatives.

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